

DE Final
Hayden Fuller

P1:

$AB > Y$

1. $00 > 1$
2. $01 > 0$
3. $10 > 0$
4. $11 > 1$

$$R = 12.5k = R_n/1 = R_p/2.4$$

a)

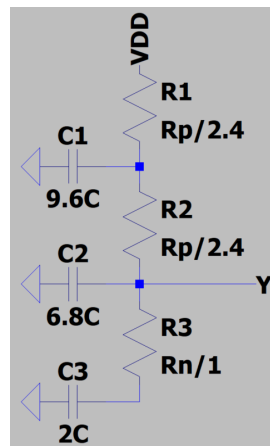
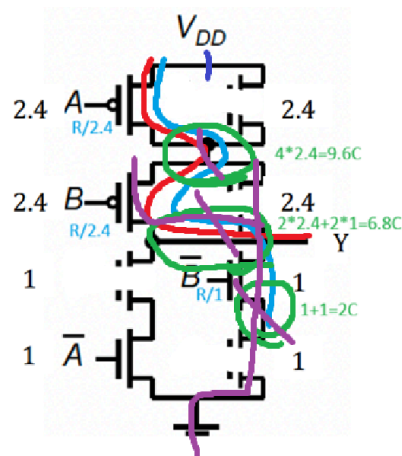
1. $10\ 0 > 00\ 1$

off off > on off

on on > on off

off on > off on

off on > on off



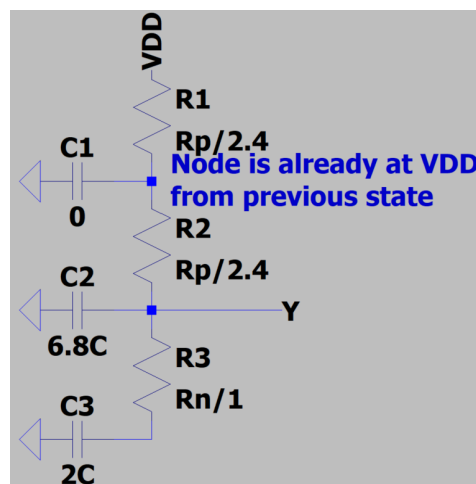
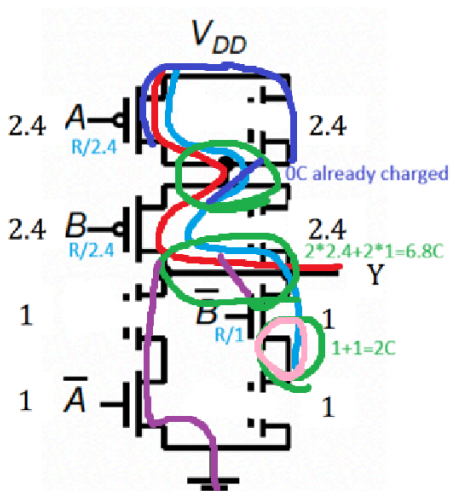
2. $01\ 0 > 00\ 1$

on on > on off

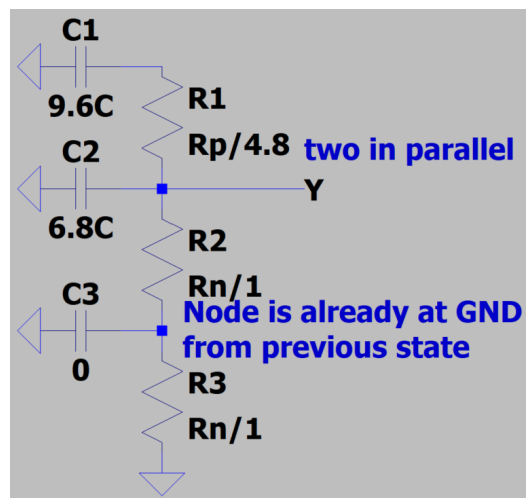
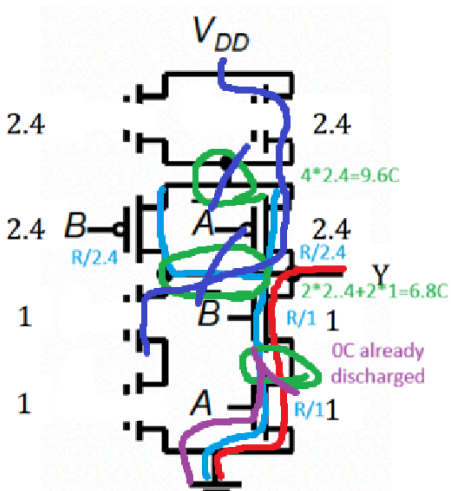
off off > on off

on off > off on

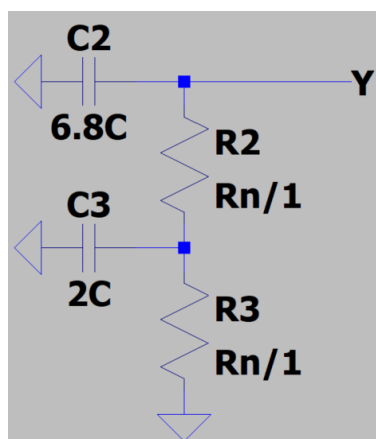
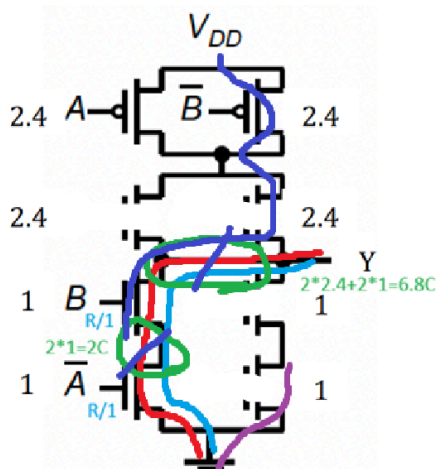
on off > on off



3. 11 1 > 10 0
 off on > off off
 off on > on on
 on off > off on
 off on > off on



4. 11 1 > 01 0
 off on > on on
 off on > off off
 on off > on off
 off on > on off



desired path
 parasitic path
 previous ground
 previous vdd
 previous floating

b)

i)

$$R = 12.5k \text{ ohm} = R_n/1 = R_p/2.4$$

$$C = W \cdot C_{self} = 1fF/\mu m \cdot 100nm = 0.1fF$$

$$RC = 12.5k \text{ ohm} \cdot 0.1fF = 1.25ps$$

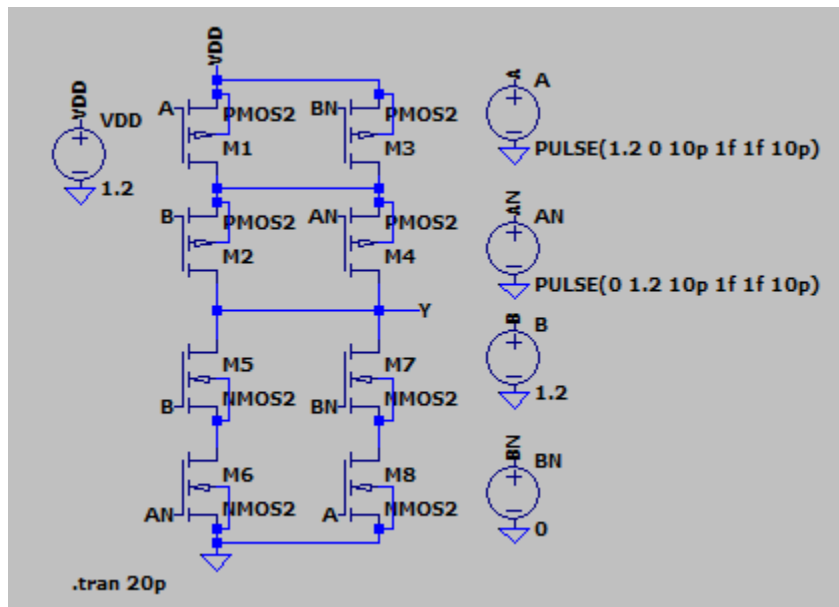
$$1. \tau = R(9.6C) + 2R(6.8C) + 2R(2C) = 27.2RC = 34ps$$

$$2. \tau = R(0) + 2R(6.8C) + 2R(2C) = 17.6RC = 22ps$$

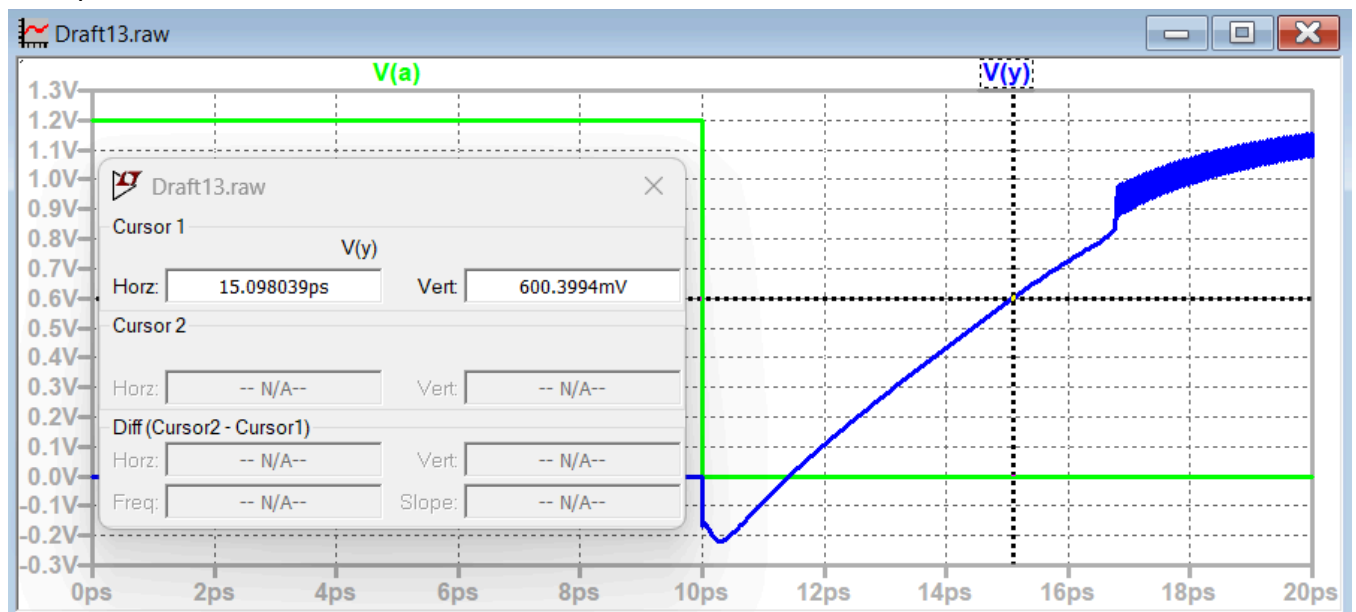
$$3. \tau = R(0) + 2R(6.8C) + 2R(9.6C) = 32.8RC = 41ps$$

$$4. \tau = R(2C) + 2R(6.8C) = 15.6RC = 19.5ps$$

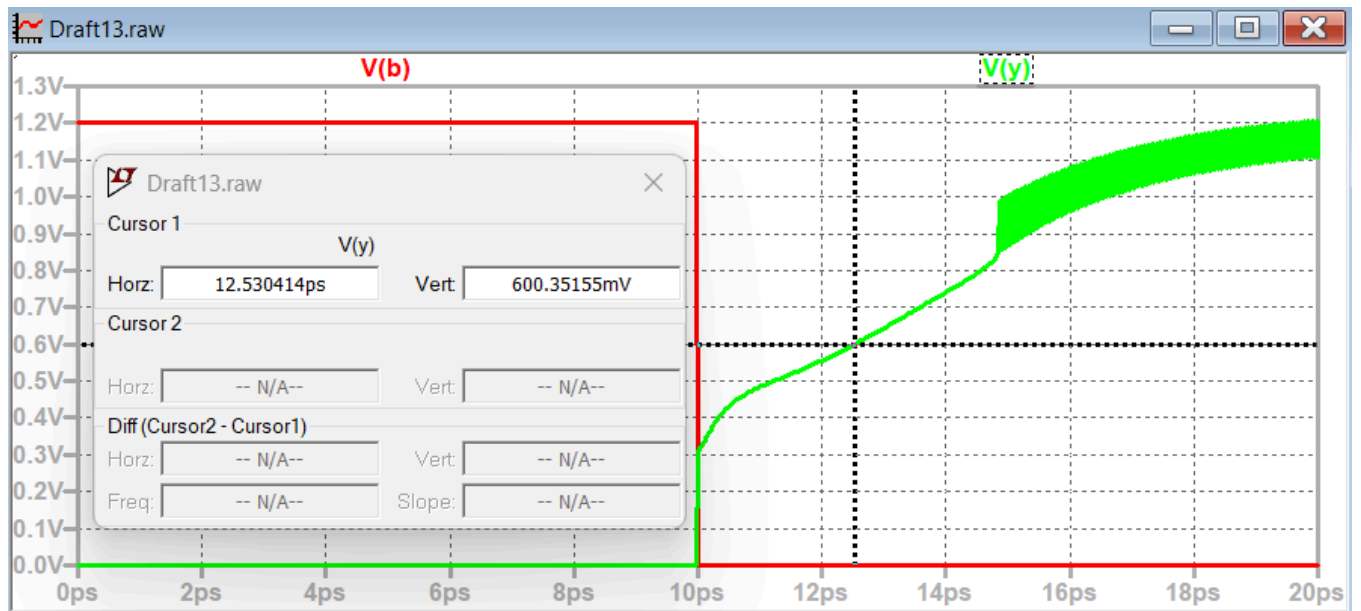
ii)



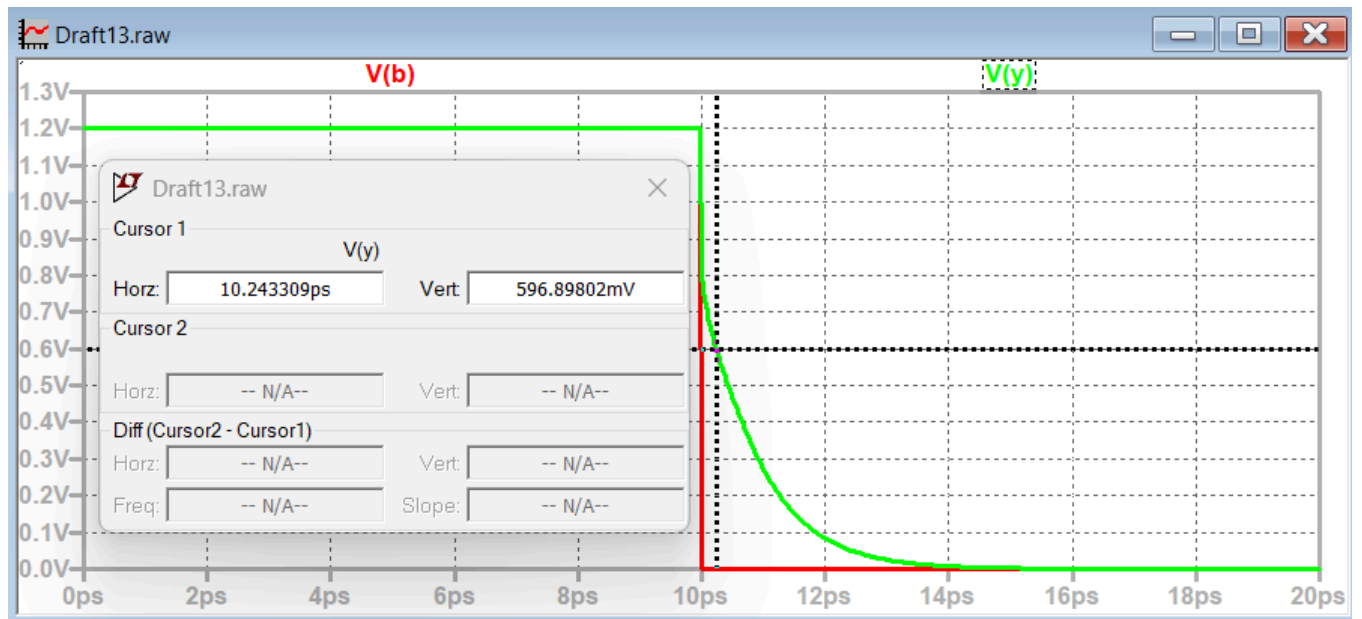
1. 5.1ps



2. 2.5ps



3. 0.2ps



My hand calculated values are higher than the LTspice simulation, which seems consistent with everything I've seen about how conservative this hand calculated approximation is.

4.

This crashes. I've spent the past hour trying to figure it out. All the other simulations work fine, this one crashes. It says that the Oxide thickness, Length, and Width are thinner than recommended for level 2 mosfet, but it's just the given 1.2E-09 and 2.2E-09, and the correct 100n and 240n. It runs for the first 10ps and the simulation hangs or crashes right at this 50% mark. I've not found any simulation parameters that fix this or any minor circuit alterations that don't change the time by orders of magnitude. Adding an inverter in the next part c moves the problem to situation 2.

```
LTspice 26.0.1 for Windows
Circuit: C:\Users\Hayden\Documents\LTspice\Draft13.net
Start Time: Thu Apr 30 17:27:55 2026
solver = Normal
Maximum thread count: 16
tnom = 27
temp = 27
method = trap
Model "nmos2": Oxide thickness thinner than recommended for a level 2 MOSFET.
Instance "M5": Length shorter than recommended for a level 2 MOSFET.
Instance "M5": Width narrower than recommended for a level 2 MOSFET.
Instance "M6": Length shorter than recommended for a level 2 MOSFET.
Instance "M6": Width narrower than recommended for a level 2 MOSFET.
Instance "M7": Length shorter than recommended for a level 2 MOSFET.
Instance "M7": Width narrower than recommended for a level 2 MOSFET.
Instance "M8": Length shorter than recommended for a level 2 MOSFET.
Instance "M8": Width narrower than recommended for a level 2 MOSFET.
Model "pmos2": Oxide thickness thinner than recommended for a level 2 MOSFET.
Instance "M1": Length shorter than recommended for a level 2 MOSFET.
Instance "M1": Width narrower than recommended for a level 2 MOSFET.
Instance "M2": Length shorter than recommended for a level 2 MOSFET.
Instance "M2": Width narrower than recommended for a level 2 MOSFET.
Instance "M3": Length shorter than recommended for a level 2 MOSFET.
Instance "M3": Width narrower than recommended for a level 2 MOSFET.
Instance "M4": Length shorter than recommended for a level 2 MOSFET.
Instance "M4": Width narrower than recommended for a level 2 MOSFET.
Direct Newton iteration succeeded in finding operating point.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0000658267487745e-11
Convergence Failure: Time step too small; time = 1.00007e-11, timestep = 2.44141e-24: trouble with node "y"

Simulation Failed: Iteration limit reached

Total elapsed time: 0.125 seconds.
```

c)

i)

$$R = 12.5 \text{ k ohm} = R_n/1 = R_p/2.4$$

$$C = W \cdot C_{\text{self}} = 1 \text{ fF/um} \cdot 100 \text{ nm} = 0.1 \text{ fF}$$

$$RC = 12.5 \text{ k ohm} \cdot 0.1 \text{ fF} = 1.25 \text{ ps}$$

$$CG = W \cdot C_{\text{gate}} = 2 \text{ fF/um} \cdot 100 \text{ nm} = 0.2 \text{ fF} = 2C$$

$$RCG = 12.5 \text{ k ohm} \cdot 0.2 \text{ fF} = 2RC = 2.5 \text{ ps}$$

Y:

across every situation adds

2 gates of 2.4W and 1W,

at 2fF/um that's $3.4 \cdot 100 \text{ nm} \cdot 2 \text{ fF/um} = .68 \text{ fF}$

always after two mosfets so $.68 \text{ fF} \cdot 2R = .68 \text{ fF} \cdot 2 \cdot 12.5 \text{ k ohm} = 17 \text{ ps}$

$$1. \tau = R(9.6C) + 2R(6.8C + 3.4CG) + 2R(2C) = 40.8RC = 51 \text{ ps}$$

$$2. \tau = R(0) + 2R(6.8C + 3.4CG) + 2R(2C) = 31.2RC = 39 \text{ ps}$$

$$3. \tau = R(0) + 2R(6.8C + 3.4CG) + 2R(9.6C) = 46.4RC = 58 \text{ ps}$$

$$4. \tau = R(2C) + 2R(6.8C + 3.4CG) = 29.2RC = 36.5 \text{ ps}$$

$$Y': \tau = R(3.4C) = 4.25 \text{ ps}$$

total for Y':

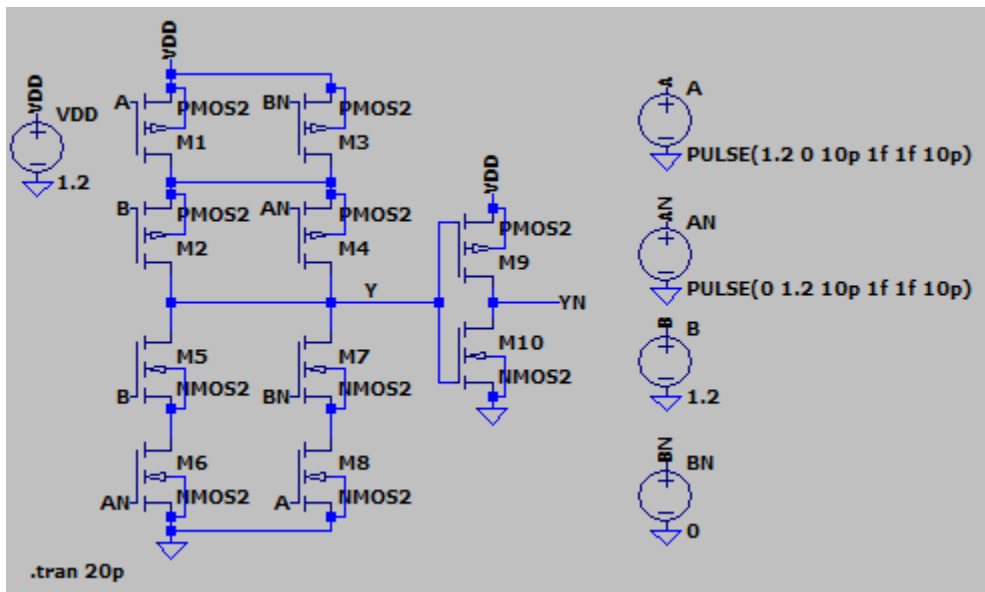
$$1. 55.25 \text{ ps}$$

$$2. 43.25 \text{ ps}$$

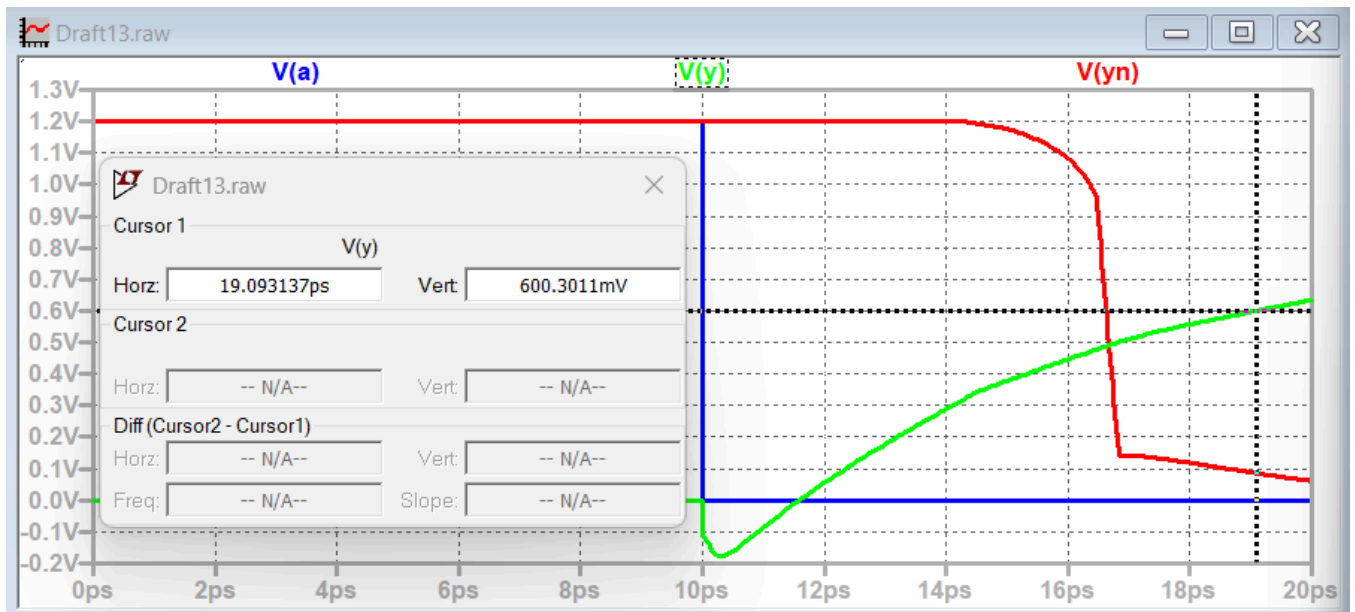
$$3. 62.25 \text{ ps}$$

$$4. 40.25 \text{ ps}$$

ii)



1. 9.1ps, 6.6ps

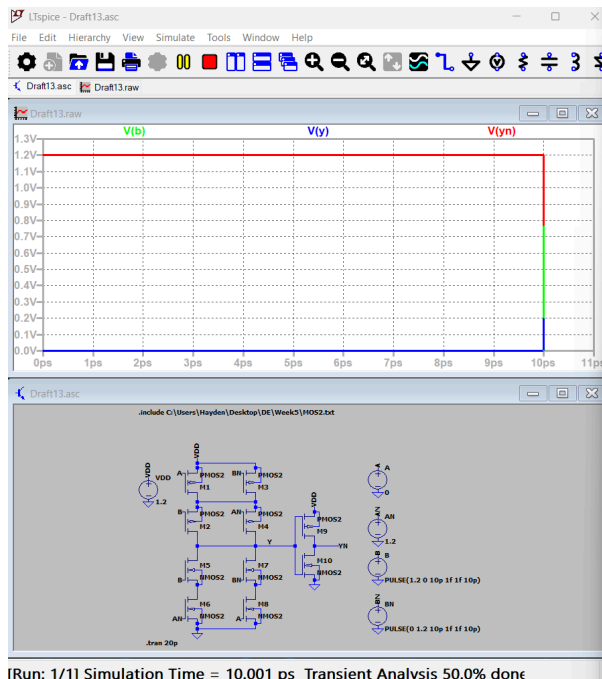


2.

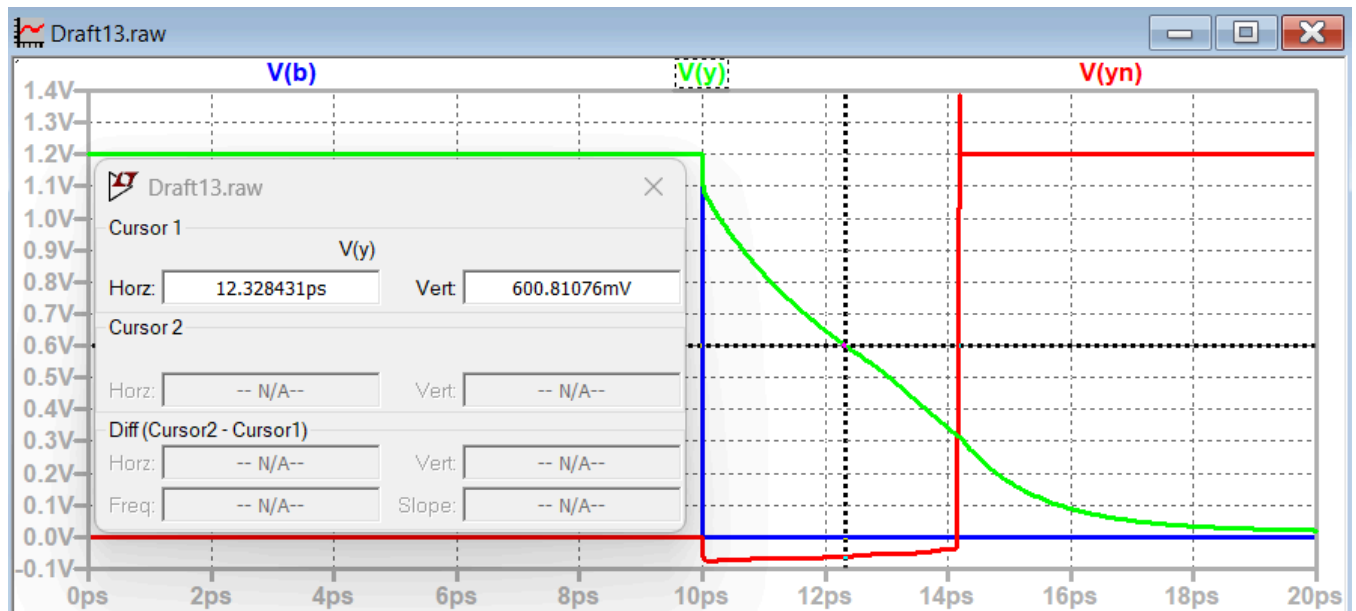
As previously mentioned in P1.a.ii.4, this crashes when the inputs flip.

```
SPICE Output Log: C:\Users\Hayden\Documents\LTspice\Draft13.log

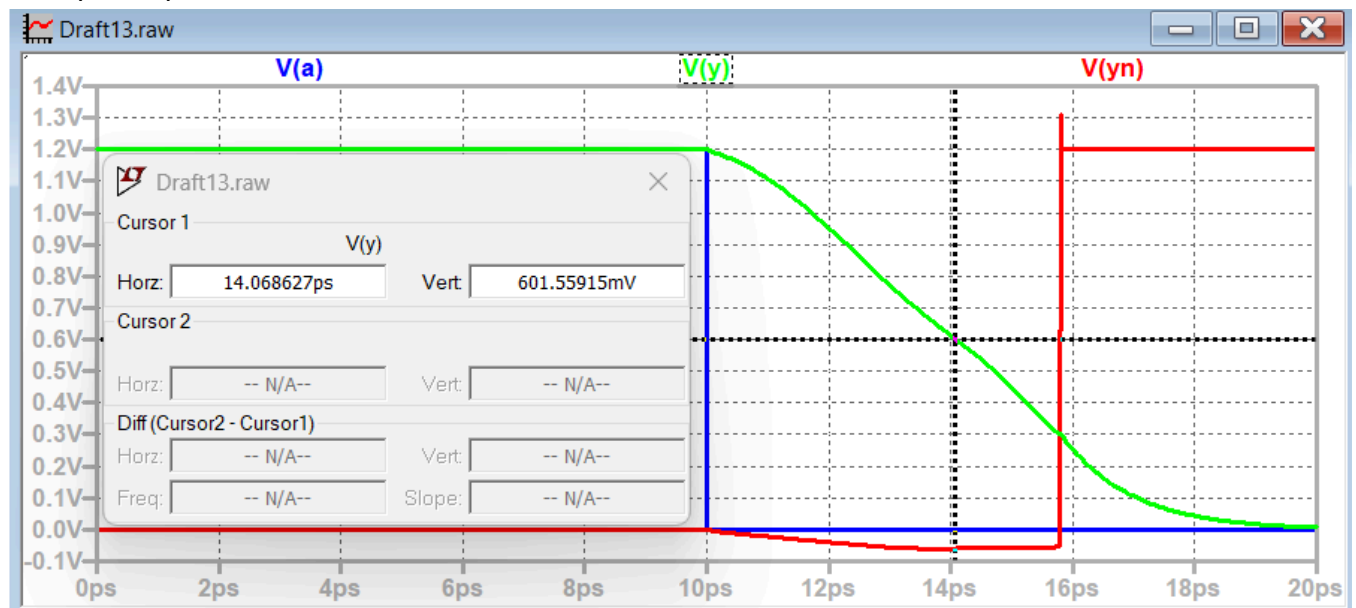
LTspice 26.0.1 for Windows
Circuit: C:\Users\Hayden\Documents\LTspice\Draft13.net
Start Time: Thu Apr 30 17:55:08 2026
solver = Normal
Maximum thread count: 16
tnom = 27
temp = 27
method = trap
Model "nmos2": Oxide thickness thinner than recommended for a level 2 MOSFET.
Instance "M5": Length shorter than recommended for a level 2 MOSFET.
Instance "M5": Width narrower than recommended for a level 2 MOSFET.
Instance "M6": Length shorter than recommended for a level 2 MOSFET.
Instance "M6": Width narrower than recommended for a level 2 MOSFET.
Instance "M7": Length shorter than recommended for a level 2 MOSFET.
Instance "M7": Width narrower than recommended for a level 2 MOSFET.
Instance "M8": Length shorter than recommended for a level 2 MOSFET.
Instance "M8": Width narrower than recommended for a level 2 MOSFET.
Instance "M10": Length shorter than recommended for a level 2 MOSFET.
Instance "M10": Width narrower than recommended for a level 2 MOSFET.
Model "pmos2": Oxide thickness thinner than recommended for a level 2 MOSFET.
Instance "M1": Length shorter than recommended for a level 2 MOSFET.
Instance "M1": Width narrower than recommended for a level 2 MOSFET.
Instance "M2": Length shorter than recommended for a level 2 MOSFET.
Instance "M2": Width narrower than recommended for a level 2 MOSFET.
Instance "M3": Length shorter than recommended for a level 2 MOSFET.
Instance "M3": Width narrower than recommended for a level 2 MOSFET.
Instance "M4": Length shorter than recommended for a level 2 MOSFET.
Instance "M4": Width narrower than recommended for a level 2 MOSFET.
Instance "M9": Length shorter than recommended for a level 2 MOSFET.
Instance "M9": Width narrower than recommended for a level 2 MOSFET.
Direct Newton iteration succeeded in finding operating point.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437539645e-11 to 1.0001005437559176e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437559176e-11 to 1.0001005437578707e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437578707e-11 to 1.0001005437598238e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437598238e-11 to 1.0001005437617768e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437617768e-11 to 1.0001005437637299e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437637299e-11 to 1.0001005437656830e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437656830e-11 to 1.0001005437676361e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437676361e-11 to 1.0001005437695892e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437695892e-11 to 1.0001005437715423e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437715423e-11 to 1.0001005437734953e-11 seconds.
Warning: Simulation tolerance relaxed to achieve convergence from 1.0001005437734953e-11 to 1.0001005437754484e-11 seconds.
```



3. 2.3ps, 4.2ps



4. 4.1ps, 5.8ps



Interestingly, in part 1 the inverter actually flips before its input reaches the 0.6V mark so its output is faster than its input. These values have a similar relation to the hand calculations as in part b,

d)

this makes no change to propagation time for Y,

1. 51ps
2. 39ps
3. 58ps
4. 36.5ps

but for Y' it adds a large capacitor on the output

$C = 10\text{mm} * 0.1\text{fF}/\mu\text{m} = 10,000 * 0.1\text{fF} = 1\text{pF}$, the old 0.34fF becomes negligible.

$12.5\text{k ohm} * 1\text{pF} = 12.5\text{ns}$

giving totals

1. 12.55525ns
2. 12.54325ns
3. 12.56225ns
4. 12.54025ns

e)

at Y this would double the gate capacitance, adding another .68fF and another 17ps to each scenario

at Y' this would halve the resistance and halve the time of the inverter, giving us about 6.25ns total

f)

this is an XNOR gate, this gives a 1 if the inputs are the same and a 0 if the inputs are different

2)

if $B > \text{not}B$, Q5 Q1 ON, Q6 Q2 OFF, output depends on A

if $\text{not}B > B$, Q6 Q2 ON, Q5 Q1 OFF, Y gets current

only matters if $B > \text{not}B$:

if $A > \text{not}A$, Q3 ON, Q4 OFF, X gets current

if $\text{not}A > A$, Q4 ON, Q3 OFF, Y gets current

current pulls output low, so inverted

A notA

B Y X

notB X X

$X = A \text{ NAND } B$

$Y = A \text{ AND } B$

a)

$RC=250$

no current = $V_{OH} = 0V$

$1mA = V_{OL} = -0.25V$

input switches at $A=\text{not}A$ and $B=\text{not}B$

$V_{ILA}=V_{IHA}=\text{not}A$

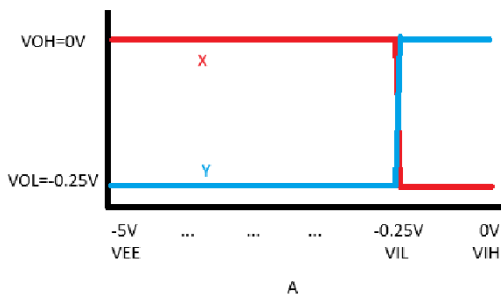
$V_{ILB}=V_{IHB}=\text{not}B$

b)

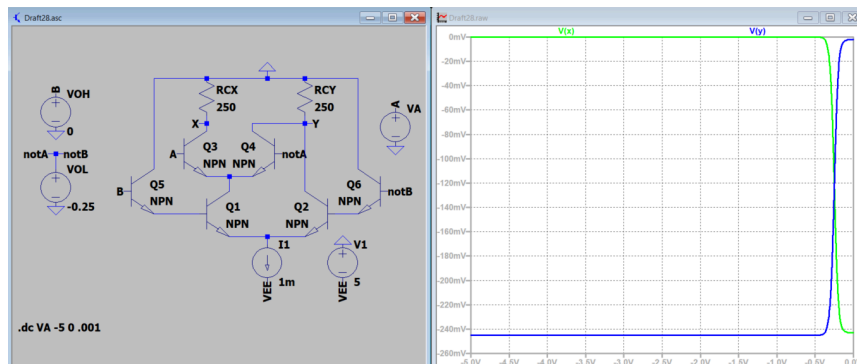
fixing $\text{not}A=\text{not}B=V_{OL}=-0.25V$, $B=V_{OH}=0V$

fixes $V_{ILA}=V_{IHA}=\text{not}A=0V$, Q1 ON, Q2 OFF, if $A > 0.25V$, Y, else, X

i)



ii)



c)

$$NMH = V_{OH} - V_{IH} = 0 - (-0.25) = 0.25V$$

$$NML = V_{IL} - V_{OL} = (-0.25) - (-0.25) = 0V$$

constant current switches to either X or Y, they have the same noise margin.

d)

as found earlier:

$$X = A \text{ NAND } B$$

$$Y = A \text{ AND } B$$

e)

decreases the load on the previous stage, a tiny current is needed by Q5/6 to give the needed current to Q1/2. With $\beta = 100$, Q1/2 need $1mA/100 = 10\mu A$. Q5/6 need to give $10\mu A$, and only require $10\mu A/100 = 100nA$. A given source can also now fill the capacitance it's driving faster and switch faster.

f)

i)

we want to drive A, so driven A and B must be high with current through X

we need X to be high to be driving, so in the driving circuit $A = -0.25$, $\text{not}A = 0$

X drawing current draws through RC, will cause X to drop

driving only one transistor at A, so $I_B = 1mA/100 = 10\mu A$

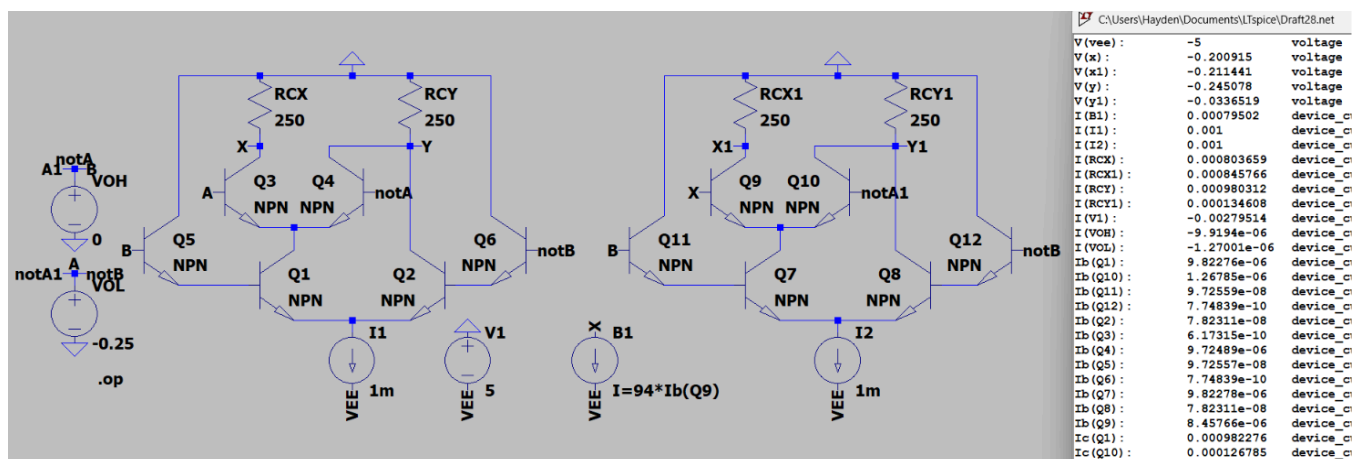
$$\Delta V = N * 10\mu A * 250 = N * 2.5mV$$

$$NMH = 0.25V$$

$$N = 100$$

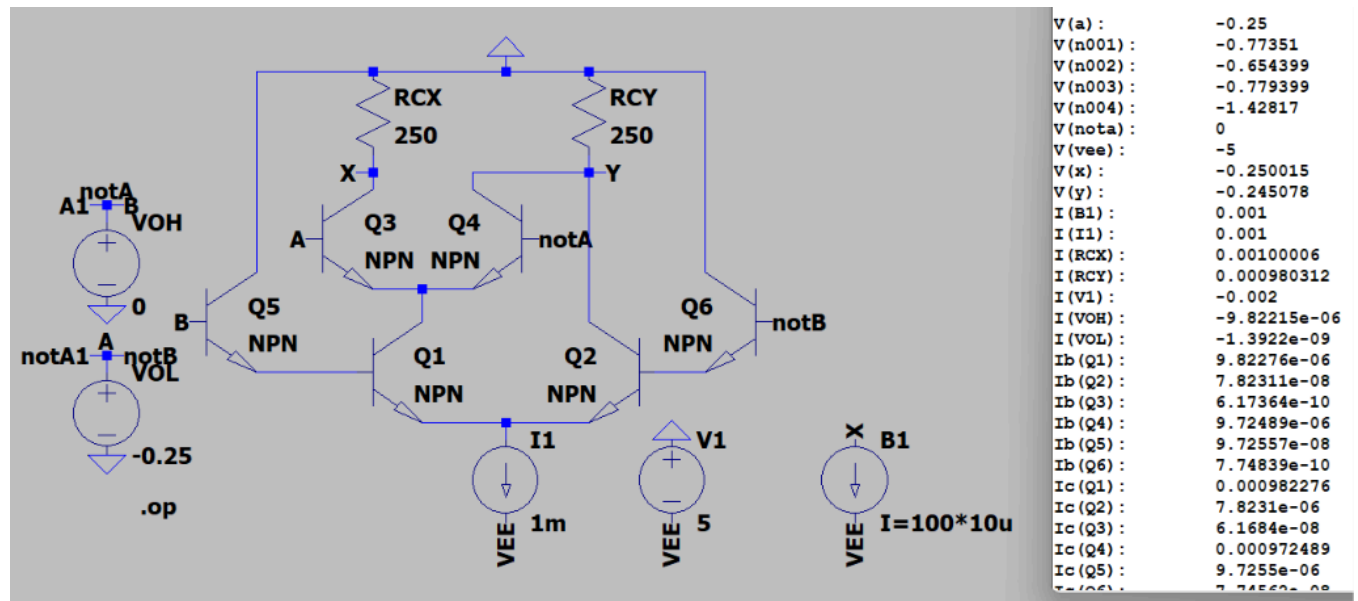
ii)

Simulating by having one duplicate circuit, then multiplying the current drawn with a dependent source
simulating $N = 95$ we get $X = -0.200915V$, as well as the output on the driven circuit rising all the way to $-0.211V$



simulating $N \geq 96$ is unable to find a solution

simplifying to our theoretical 10uA each, we get up to 99, just barely going under -0.25V at 100.



g)

i)

this drops X and Y by a constant 0.7V,

VOH=-0.7V

VOL=-0.95V

ii)

now output current is provided by a transistor, the old X node only needs to supply 1/100 of its old current + 1m,

$I_B = (1m + N \cdot 10u) / 100$

$0.25V / 250 = 1mA = I_{Bmax}$

$1m \geq (1m + N \cdot 10u) / 100$

$N = 9900$

3)

a)

writing $Q=1$ not $Q=0$

PC high, T7 T8 OFF

$W=1$ not $W=0$, T10 ON, T9 OFF

$WL=1$, T5 T6 ON

T1 T4 help pull Q HI not Q LO

T2 T3 fight against pulling Q LO not Q HI

T10 T5 T6 T2 T1 T2 T3 T4

b)

assuming we're still in the end of the write stage with $WL=HI$ etc

$Q=1.2V$, not $Q=0V$

cutoff: $V_{GS} < V_T$, $I_D = 0$

linear: $V_{GS} \geq V_T$ and $V_{DS} \leq V_{GS} - V_T$, $I_D = \frac{k}{2} [2(V_{GS} - V_T) V_{DS} - V_{DS}^2]$

saturation: $V_{GS} \geq V_T$ and $V_{DS} \geq V_{GS} - V_T$, $I_D = \frac{k}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

T1: linear

$V_{SG} = 1.2 > 0.4 = V_T$ pass

$V_{SD} = 0 < 0.8 = 1.2 - 0.4 = V_{SG} - V_T$ fail

T2: cutoff

$V_{SG} = 0 < 0.4 = V_T$ fail

T3: cutoff

$V_{GS} = 0 < 0.4 = V_T$ fail

T4: linear

$V_{GS} = 1.2 > 0.4 = V_T$ pass

$V_{DS} = 0 < 0.8 = 1.2 - 0.4 = V_{GS} - V_T$ fail

T5: cutoff

$V_{GS} = 0 < 0.4 = V_T$ fail

T6: linear

$V_{GS} = 1.2 > 0.4 = V_T$ pass

$V_{DS} = 0 < 0.8 = 1.2 - 0.4 = V_{GS} - V_T$ fail

T7: cutoff

$V_{SG} = 0 < 0.4 = V_T$ fail

T8: cutoff

$V_{SG} = 0 < 0.4 = V_T$ fail

T9: cutoff

$V_{GS} = 0 < 0.4 = V_T$ fail

T10: linear

$V_{GS} = 1.2 > 0.4 = V_T$ pass

$V_{DS} = 0 < 0.8 = 1.2 - 0.4 = V_{GS} - V_T$ fail

on linear: T1 T4 T6 T10

off cutoff: T2 T3 T5 T7 T8 T9

c)

i)

assume Q and notQ pulled most of the way to VDD/2, BL and notBL still close to VDD and 0

$Q \sim 0.6V$, $\text{not}Q \sim 0.6V$

cutoff: $V_{GS} < V_T$, $I_D = 0$

linear: $V_{GS} \geq V_T$ and $V_{DS} \leq V_{GS} - V_T$, $I_D = \frac{k}{2} [2(V_{GS} - V_T) V_{DS} - V_{DS}^2]$

saturation: $V_{GS} \geq V_T$ and $V_{DS} \geq V_{GS} - V_T$, $I_D = \frac{k}{2} (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$

T1: saturation

$V_{SG} \sim 0.6 > 0.4 = V_T$ pass

$V_{SD} \sim 0.6 > 0.2 = 0.6 - 0.4 = V_{SG} - V_T$ pass

T2: saturation

$V_{SG} \sim 0.6 > 0.4 = V_T$ pass

$V_{SD} \sim 0.6 > 0.2 = 0.6 - 0.4 = V_{SG} - V_T$ pass

T3: saturation

$V_{GS} \sim 0.6 > 0.4 = V_T$ pass

$V_{DS} \sim 0.6 > 0.2 = 0.6 - 0.4 = V_{GS} - V_T$ pass

T4: saturation

$V_{GS} \sim 0.6 > 0.4 = V_T$ pass

$V_{DS} \sim 0.6 > 0.2 = 0.6 - 0.4 = V_{GS} - V_T$ pass

T5: saturation

$V_{GS} \sim 0.6 > 0.4 = V_T$ pass

$V_{DS} \sim 0.6 > 0.2 = 0.6 - 0.4 = V_{GS} - V_T$ pass

T6: linear

$V_{GS} \sim 0.6 > 0.4 = V_T$ pass

$V_{DS} = -0.6 < 0.2 = 0.6 - 0.4 = V_{GS} - V_T$ fail

T7: cutoff

$V_{SG} = 0 < 0.4 = V_T$ fail

T8: cutoff

$V_{SG} = 0 < 0.4 = V_T$ fail

T9: cutoff

$V_{GS} = 0 < 0.4 = V_T$ fail

T10: linear

$V_{GS} = 1.2 > 0.4 = V_T$ pass

$V_{DS} = 0 < 0.8 = 1.2 - 0.4 = V_{GS} - V_T$ fail

on saturation: T1 T2 T3 T4 T5

on linear: T6 T10

off cutoff: T7 T8 T9

ii)

current into notQ through T2, out through T4||T6, notBL=0 so assume N10 is negligible

$$IT2(\text{saturation}) = IT4(\text{saturation}) + IT6(\text{linear})$$

$$IT2(\text{saturation}) = k_p/2 (V_{SG2} - V_T)^2 (1 + \lambda V_{SD2})$$

$$V_{SG2} = (1.2 - Q), V_{SD2} = (1.2 - \text{not}Q), k_p = 70 \times 2.4 = 168$$

$$IT2(\text{saturation}) = 84 ((1.2 - Q) - 0.4)^2 (1 + 0.2(1.2 - \text{not}Q))$$

$$IT2(\text{saturation}) = 84 (0.8 - Q)^2 (1.24 - 0.2\text{not}Q)$$

$$IT2(\text{saturation}) = 84 (Q^2 - 1.6Q + 0.64) (-0.2\text{not}Q + 1.24)$$

$$IT4(\text{saturation}) = k_n/2 (V_{GS4} - V_T)^2 (1 + \lambda V_{DS4})$$

$$V_{GS4} = Q, V_{DS4} = \text{not}Q, k_n = 270 \times 1$$

$$IT4(\text{saturation}) = 135 (Q - 0.4)^2 (1 + 0.2\text{not}Q)$$

$$IT4(\text{saturation}) = 135 (Q^2 - 0.8Q + 0.16) (0.2\text{not}Q + 1)$$

$$IT6(\text{linear}) = k/2 [2(V_{GS6} - V_T) V_{DS6} - V_{DS6}^2]$$

$$V_{GS6} = (1.2 - \text{not}Q), V_{DS6} = (-\text{not}Q), k_n = 270 \times 1$$

$$IT6(\text{linear}) = 135 [2((1.2 - \text{not}Q) - 0.4) (-\text{not}Q) - (-\text{not}Q)^2]$$

$$IT6(\text{linear}) = 135 [2(0.8 - \text{not}Q) (-\text{not}Q) - \text{not}Q^2]$$

$$IT6(\text{linear}) = 135 (\text{not}Q^2 - 1.6\text{not}Q)$$

$$84 (Q^2 - 1.6Q + 0.64) (-0.2\text{not}Q + 1.24) = 135 (Q^2 - 0.8Q + 0.16) (0.2\text{not}Q + 1) + 135 (\text{not}Q^2 - 1.6\text{not}Q)$$

current into Q through T1, out through T3, BL is just capacitance so ignore T5

$$IT1(\text{saturation}) = IT3(\text{saturation})$$

$$IT1(\text{saturation}) = k_p/2 (V_{SG1} - V_T)^2 (1 + \lambda V_{SD1})$$

$$V_{SG1} = (1.2 - \text{not}Q), V_{SD1} = (1.2 - Q), k_p = 168$$

$$IT1(\text{saturation}) = 84 ((1.2 - \text{not}Q) - 0.4)^2 (1 + 0.2(1.2 - Q))$$

$$IT1(\text{saturation}) = 84 (-0.2Q + 1.24) (\text{not}Q^2 - 1.6\text{not}Q + 0.64)$$

$$IT3(\text{saturation}) = k_n/2 (V_{GS3} - V_T)^2 (1 + \lambda V_{DS3})$$

$$V_{GS3} = \text{not}Q, V_{DS3} = Q, k_n = 270 \times 1$$

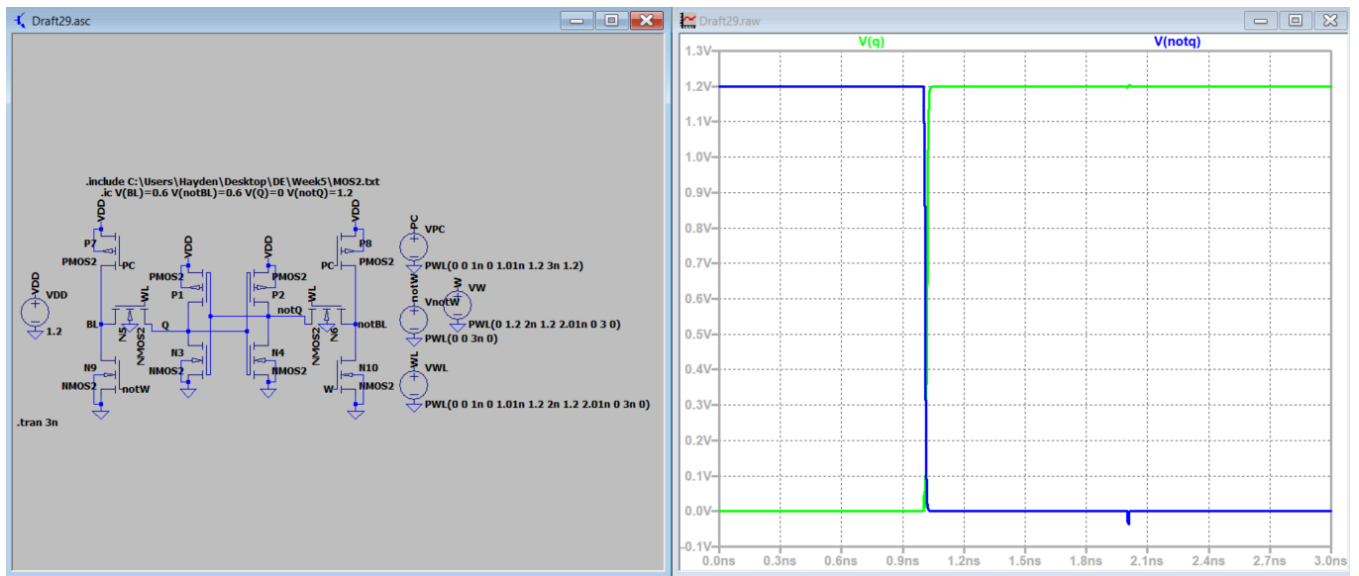
$$IT3(\text{saturation}) = 135 (0.2Q + 1) (\text{not}Q^2 - 0.8\text{not}Q + 0.16)$$

$$84 (-0.2Q + 1.24) (\text{not}Q^2 - 1.6\text{not}Q + 0.64) = 135 (0.2Q + 1) (\text{not}Q^2 - 0.8\text{not}Q + 0.16)$$

two solutions: notQ=0.566076, Q=1.18766 OR notQ=0.619707, Q=-1.69289

notQ=0.566076, Q=1.18766, so it just successfully wrote...

iii + iiii)



LTspice simulates a successful write, just like my calculations found. This circuit writes quite easily, in order to purposefully make it fail, I had to drop T10 to W=100n, as well as increase T1 T2 W=370n.